

Transforming FMCG Marketing with AI-Powered Predictive Analytics: Insights from Multi-Model Evaluation

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Abstract:

In the Fast-Moving Consumer Goods (FMCG) industry which is characterized by high level of competition, fluctuating consumer demand, and very thin profit margins success is a result of being able to predict market trends instead of just reacting to them. This case study looks at the key role of Predictive Analytics (PA) which is supported by Artificial Intelligence (AI) and Machine Learning (ML) in terms of it being a trigger for marketing innovation and strategic decision making within the FMCG sector. We present a practical example of a multi algorithmic AI system we developed for sales volume prediction which is a very important element of marketing return on investment, supply chain performance and total profit. Our approach is to systematize the training and evaluation of a set of regression models which include Linear Regression, Ridge Regression, Support Vector Regression (SVR), Random Forest, and Gradient Boosting on a large scale FMCG data which we include temporal and promotional variables in. We partition the data into train and test sets which in turn we use for in depth validation of the predictive models. We use key performance indicators like Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) to very closely compare model performance. What we find from our study is that ensemble methods in particular Gradient Boosting and Random Forest do very well in outperforming traditional linear and kernel-based models at what they do which is to put forward complex non-linear associations between many marketing drivers and sales results. With advanced predictive tools, marketing is changing from guesswork to smart strategies based on data. This shift allows companies to engage with customers more personally by targeting them more accurately. It also helps businesses improve their trade promotions by predicting sales boosts and guiding decisions about pricing and launching new products.

Additionally, using these predictions within inventory management systems can cut down on waste throughout the supply chain. This study shows that using strong AI-driven predictive analytics isn't just a nice-to-have anymore; it's essential for fast-moving consumer goods (FMCG) companies to stay competitive, adapt to market changes, and grow sustainably in today's digital world. The case study offered is a helpful roadmap for other businesses looking to implement similar AI solutions effectively.

Keywords: Predictive Analytics, Artificial Intelligence, Machine Learning, FMCG, Demand Forecasting, Marketing Innovation, Ensemble Methods, Support Vector Regression, Gradient Boosting, Random Forest.

Introduction:

You know, entering into the world of Fast-Moving Consumer Goods (FMCG) is like jumping into a blender – it's just so dynamic and incredibly competitive. We're talking about all those things you grab without a second thought: from your morning coffee, your toothpaste, the cleaning stuff under the sink to all the human necessary things. These products literally fly off the shelves! Sure, there's a ton of money changing hands, but the profit margins? Often super slim. Sales volumes are through the roof, and what people want seems to change almost daily. This constant whirl, even though it's profitable, brings a whole lot of headaches: brutal competition, demand that's anyone's guess, the relentless need for fresh ideas, unexpected market curveballs, and the struggle to keep supply chains and inventory perfectly in sync is never-ending. One of the absolute biggest headaches for FMCG companies is trying to figure out what folks are actually going to buy. Seriously, getting demand right is the secret sauce for pretty much every big business decision – whether you're mapping out a huge marketing blitz, launching special offers, managing your stock, sorting out deliveries, or even rolling out a brand-new product.

The thing is, those old-school ways of forecasting, like just glancing at past averages or simple trend lines, just can't keep pace with today's super complex, nuanced market. And when those predictions are off? Oh man, it costs a fortune: too much product sitting around gathering dust, empty shelves when customers are desperate for something, missing out on sales, throwing money away on promotions that flop, and ultimately, a direct hit to the bottom line.

But here's where it gets really exciting: AI and Machine Learning are completely turning this industry on its head. Imagine AI tools that can smoothly sift through mountains of data – not just old sales numbers, but pricing strategies, on what promotions are running, what competitors are planning, and even outside stuff like the economy or the weather. These sophisticated analytics can spot hidden patterns and tricky connections that traditional methods would totally miss, shifting demand forecasting from a hopeful guess to a precise, data-backed science and that's exactly what we've been digging into with our research. We put together a real-world case study to show how a mix of AI models can nail sales volume predictions in the FMCG world. We used this extremely detailed dataset, with all sorts of info like dates, product types, prices, promotions, store locations, and even the nitty-gritty of the supply chain.

To get a really clear, we didn't just pick one model; we threw several into the ring – from simpler ones like Linear and Ridge Regression to the more complex beasts like Support Vector Regression (SVR), Random Forest, and Gradient Boosting. This multi-model approach not only gave us incredibly accurate predictions but also insights we could actually use in the business. And naturally, we rigorously checked how well each model performed using the usual suspects: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2).

Our paper really aimed to do three main things:

First, we wanted to truly get under the hood and compare different AI models – from the straightforward linear ones to the more advanced ensemble methods – to see which ones the truly champions are by using a robust, real-world dataset for forecasting FMCG sales.

Second, it wasn't just about being technically accurate. We really wanted to show how you can take those model outputs and turn them into genuine business value. Think about things like getting a much bigger bang for your buck from marketing investments, making promotions actually work, and having a supply chain that can pivot on a dime.

Finally, we're hoping this offers a practical roadmap for anyone in FMCG who's looking to jump into AI. It's not just about getting super spot-on predictions, but about using AI as that essential tool for making smarter, faster decisions in an industry that simply never stops.

What we found was honestly pretty mind-blowing: those ensemble methods, especially Gradient Boosting

and Random Forest are absolutely best. They were miles better than traditional methods at understanding the complex, subtle ways that marketing efforts actually sway sales. Using these advanced AI models led to a really noticeable leap in forecast accuracy, giving marketers and planners the power to move from just throwing darts in the dark to genuinely data-driven strategies. On top of that, encircling these systems are away to even more cool stuff down the line, like more personalized customer experiences, squeezing every last drop of value from promotional budgets all along the FMCG supply chain.

Literature Review:

Hybrid Models for Macro-Economic Forecasting

In 2025, Nagesh et al. predicted India's FMCG market index using a long-term time-series dataset (1993-2023). The three models evaluated were a standard ARIMA model that allows the researcher to quantify seasonal and linear trends, a nonlinear support vector machine (SVM) model, and the hybrid ARIMA-SVM model which allowed researchers to combine parts of both models. The researchers described a clear hierarchy in terms of performance. For example, the ARIMA model performed better than the standalone SVM model while the hybrid ARIMA-SVM model outperformed both the linear and nonlinear models. This suggests drawing from the advantages of combining linear modeling and nonlinear modeling approaches allows for more complete tools for macroeconomic forecasting and creates benefits for decision-makers.

Deep Learning for Operational Sales and Demand Forecasting

Thejovathi and Rao tackled the challenge of franchise sales forecasting in a separate study in 2025 by reframing the problem from an FMCG perspective as a sequence problem and could be solved via LSTM networks. They set out a data processing protocols to be able to use large farm-based retail grocery data through stream processing, because of computational and feature engineering limits, by slicing data into "chunks" that represented focused "normal" operational periods (effectively a manageable historical forecast horizon). After processing the large retail data via LSTM model training, the LSTM model was able to independently and effectively identify complex dependencies, while synthesizing long-distance dependencies i.e., consumer purchasing behavior. The study identified predictions to be "very accurate", with notable; loss curve evaluations using multiple layers of local structure and feature relevance analyses. Practically, this results in improvement to how supply chains are managed and enabling further sophistication of the inventory allocation, logistics and marketing strategies which in turn reduces waste and improves operational responsiveness.

Leveraging Simple Machine Learning for Classification

Demand forecasting is approached differently in a recent work that treats the issue as a supervised classification problem (Mebal et al., 2021, IJEAT). Assuming the authors approach their work correctly, the authors used a K-Nearest Neighbors (KNN) algorithm applied to an FMCG sales dataset. After a standard training to testing split of

75%/25%, the authors achieved a whopping 99% of the time in training and testing datasets. The study's findings demonstrate that a basic machine learning model, known as KNN, can learn from past sales data and product attributes to learn and anticipate sales. The authors put an emphasis on how predictive analytics is readily available and can be used to optimize inventory and other areas of profitability, especially when aligned with forecasting demand.

A Machine Learning Approach to Inventory Stockout Prediction (Yang Liu et al., 2025)

Liu et al. address inventory optimization by predicting stockouts in a large retail dataset (over 1.6 million SKUs). They build a classification model using classical ML algorithms (e.g. tree-based ensembles) on features including current inventory levels, historical sales, and demand forecasts. Their results show the model handles extremely imbalanced, large-scale data and markedly improves stockout prediction accuracy. Crucially, their feature-importance analysis reveals that current inventory, short-term (3-month) demand forecasts and recent sales volumes are the strongest predictors of an upcoming stockout. The importance of near-term indicators for FMCG inventory management was highlighted by the significantly lower impact of longer-range forecasts (6–9 months). So, Liu et al. exhibited that furthermore forecasting out of stock products, machine learning can also identify which demand signals are most important for inventory planning.

Methodology:

Data Collection: A multi-year dataset was encircling daily transactions in the FMCG sector, includes categories such as Household, Personal Care, Dairy, Beverages, and Snacks. Each record includes critical variables:

- Date (transaction timestamp)
- Product_Category
- Sales_Volume (target/output variable)
- Price
- Promotion (binary)
- Store_Location (Urban/Suburban/Rural)
- Weekday
- Supplier_Cost
- Replenishment_Lead_Time
- Stock_Level

This collection stage ensures diverse input to drive all subsequent analysis and modeling steps.

Data Cleaning and Preprocessing:

In this stage, as shown in the flowchart, it is fundamental for robust analytics and the Steps included are:

- **Missing Value Handling:** Imputation using median (numerics) or mode (categories)
- **Outlier Treatment:** Capping extreme numerics using IQR-based rules
- **Normalization:** Standardizing features (e.g., z-score normalization) to enable fair model comparison
- **Encoding:** Categorical fields such as product type and location were one-hot encoded so that algorithms could leverage all signals

- **Data Integrity Checks:** Invalid (e.g., negative) sales costs were corrected or removed

These actions turned raw data into model-ready, reliable input.

Exploratory Data Analysis (EDA):

EDA involved orderly investigating the structured data to uncover business patterns and model signals:

• Trend and Pattern Identification:

Visualizations and descriptive statistics for variables like sales over time, seasonal effects, or promotion impact

- **Correlation Analysis:** Quantifying relationships between features (e.g., does price reduction during promotions consistently affect sales more in Urban locations?)

Also, these findings inform both feature selection and the need for interaction effects.

Feature Engineering:

In this stage, the domain knowledge and EDA patterns were used to enhance the dataset:

- **Derived Features:** Extracted components from “Date” (month, quarter, or holiday indicators); created “Is Weekend”; engineered “Promotion_Effect_Weeks” to model lagged uplift

- **Refined Categorical:** Extended encoding for hierarchical groupings if required (e.g., grouping beverage types)

- **Interaction Terms:** Combined variables (e.g., Promotion × Store Location) where EDA showed conditional effects

Strong feature engineering here directly supports improved predictive ability and actionable insights.

Model selection:

Linear regression: Linear regression represents a basic statistical learning technique that examines the relationship between dependent and independent variables by a linear equation. This method assumes a linear association between predictor variables and the target outcome, making it highly interpretable for business applications in demand forecasting.

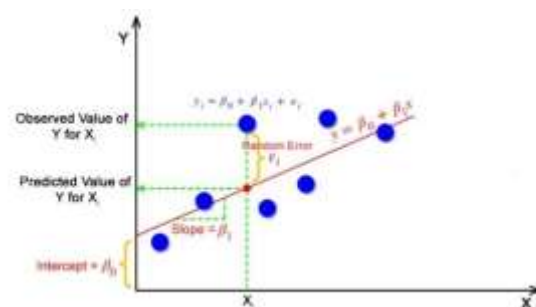


Fig1. Linear regression

Mathematical Formulation:

The simple linear regression model for a single variable predictor follows the equation:

$$Y = \beta_0 + \beta x + \varepsilon$$

where y represents the dependent variable (sales volume), x denotes the independent variable (predictor), β_0 is the y-intercept, β_1 is the slope coefficient, and ϵ represents the error term capturing unexplained variance.

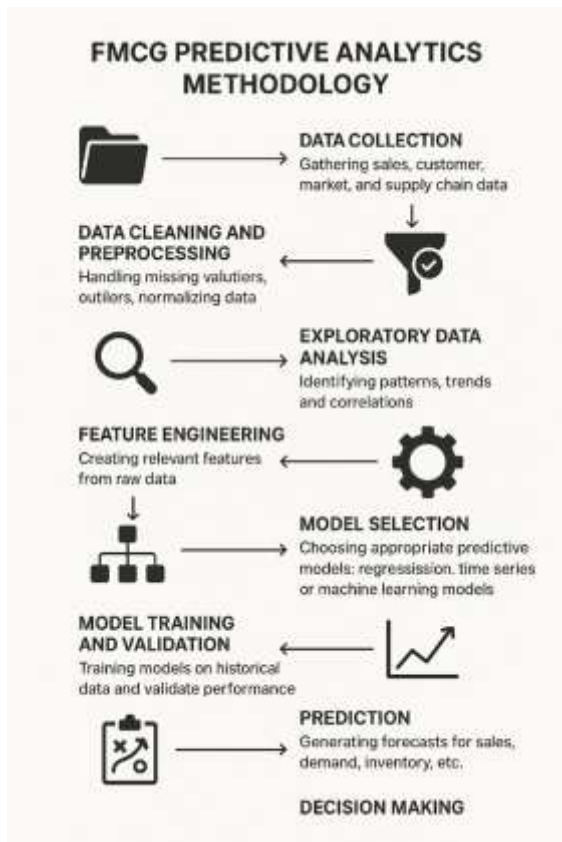


Fig2: Flow chart

Ridge Regression

Ridge regression is an extension of regularized ordinary least squares (OLS) that adds an L_2 penalty on coefficient magnitudes to reduce overfitting, especially when predictors are highly linear or when number of features (p) approaches n (number of observations). The core objective is to shrink coefficients toward zero, trading a small increase in bias for a potentially large reduction in variance and improved predictive performance.

Objective Function

Given data $\{(x_i, y_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^p$ and $y_i \in \mathbb{R}$, ridge regression minimizes:

$$L(\beta) = \sum_{i=1}^n (y_i - \beta^T x_i)^2 + \lambda \|\beta\|_2^2, L(\beta) = \frac{1}{2} \sum_{i=1}^n (y_i - \beta^T x_i)^2 + \frac{\lambda}{2} \|\beta\|_2^2$$

where $\|\beta\|_2^2 = \sum_{j=1}^p \beta_j^2$ and $\lambda \geq 0$ controls the strength of regularization.

Closed-Form Solution

In matrix notation, with $X \in \mathbb{R}^{n \times p}$ and $y \in \mathbb{R}^n$, the ridge estimator is:

$$\begin{aligned} \hat{\beta}^{\text{ridge}} &= \arg \min_{\beta} \{ (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta \} \\ &= (X^T X + \lambda I_p)^{-1} X^T y \\ &= \arg \min_{\beta} \{ (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta \} = (X^T X + \lambda I_p)^{-1} X^T y \end{aligned}$$

Support Vector Regression:

In Support vector machine (SVM), Support vector regression (SVR) is a type that is used for regression tasks. It tries to find a function that predicts the best continuous output insights for a given input value. SVR uses both linear and non-linear kernels. A linear kernel performs a simple dot product between two input vectors, while a non-linear kernel performs more complex functions that can capture more complicated and detailed patterns in the data. The choice of kernel depends on the data's characteristics and the task's complexity.

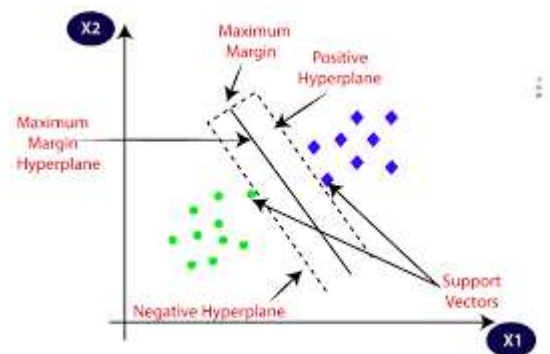


Fig3: SVR

Mathematical Formulation

Single-equation representation of the regression function with kernel K :

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha^*) K(x_i, x) + b$$

Random Forest:

Random Forest is an associated learning algorithm that builds on multiple random decision trees and groups their forecasts to enhance overall accuracy and robustness output patterns. It operates by building each tree on a bootstrap sample drawn with replacement from the training dataset, introducing randomness which increases model variations.

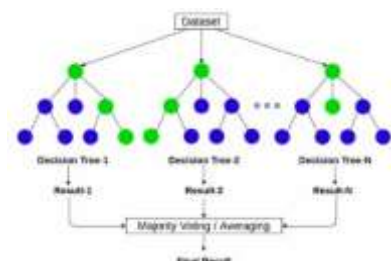


Fig 4: Random Forest

At every node in a tree, a random subset of features is selected to significant the best split, reducing correlation among trees and improving generalization.

Mathematically Formula:

Given a training set $D_n = \{(X_i, Y_i)\} \quad I = 1n$ where $X_i \in \mathbb{R}^p$ denotes the predictor vector, and Y_i the target, Random Forest fits M randomized regression trees $mn(x; \Theta_j, D_n)$, where Θ_j represents the random inputs used for resampling and feature selection in tree j . The prediction at a query point x is the average of the predictions of the individual trees:

$$m \wedge M, n(x) = \frac{1}{M} \sum_{j=1}^M mn(x; \Theta_j, D_n)$$

Gradient Boosting:

Gradient Boosting is an ensemble learning method used for classification and regression tasks. It is a boosting algorithm which combine multiple weak learners to create a strong predictive model. It works by sequentially training models where each new model tries to correct the errors made by its predecessor.

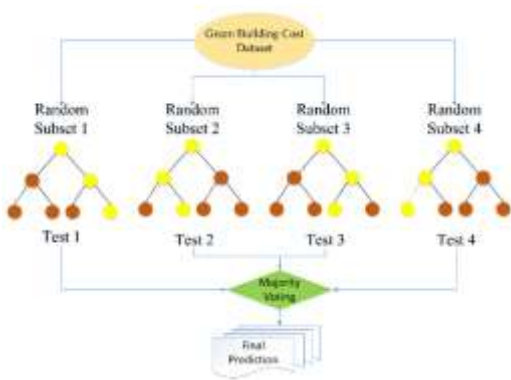


Fig5: Gradient Boosting

In gradient boosting each new model is trained to minimize the loss function such as mean squared error or cross-entropy of the previous model using gradient descent. In each iteration the algorithm computes the gradient of the loss function with respect to predictions and then trains a new weak model to minimize this gradient. Predictions of the new model are then added to the ensemble (all models prediction) and the process is repeated until a stopping criterion is met.

Mathematical Formulation

The general formula for updating the prediction in gradient boosting at each iteration m is:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x)$$

Where,

- $F_m(x)$ is the boosted model's prediction at iteration m
- F_{m-1} is the boosted model's prediction from the previous iteration ($m-1$).

Model Training and Validation:

In this step, Models were developed and validated as below:

- **Splitting:** The data is splitted into 80% of training data and 20% of testing data for independent testing.

- **Training:** In this stage it trains the model based on proposed data and feature-selected data.

- **Cross-validation:** Five-fold CV minimized bias and variance, ensuring robust performance and evaluation across models

Metrics:

RMSE:

Root Mean Square Error (RMSE) is a standard metric formula for evaluating regression models accuracy by analyzing the average magnitude of prediction errors in the same units as the target variable. For a set of n observations with all true values y_i and predicted values $\bar{y}_i$, RMSE is defined as below:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2}$$

Primarily this formula computes the squared errors $\sum_{i=1}^n (y_i - \bar{y}_i)^2$ to penalize larger deviations more heavily, averages them to obtain the Mean Squared Error (MSE), and then takes the square root to restore the original unit scale. A lower RMSE indicates that the model's predictions are, on average, closer to the actual values, making it a particularly intuitive measure for comparing models in contexts such as sales volume forecasting.

MAE: Mean Absolute Error (MAE) is a usual metric used to calculate the accuracy of regression models by measuring the average absolute difference between the predicted values and the actual values. It indicates the mean magnitude of errors without considering their direction, thus providing a linear score where all errors contribute equally.

The mathematical representation of MAE:

$$MAE = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{y}_i)$$

Where,

- n is the total number of observations,
- y_i is the actual value of observation,
- $\bar{y}_i$ is the predicted value observation.

This metric is resilient for outliers when compared to squared-error-based metrics, as it does not disproportionately penalize large errors. MAE is expressed in the same way as the target variable, making it intuitively interpretable for practical forecasting tasks such as FMCG sales prediction. A lower MAE indicates better predictive accuracy

R²:

R-squared tells you what percentage of the changes or "variance" in what you're trying to predict the "dependent variable" can be determined by the factors you're using to make that prediction (the "independent variable(s)"). It essentially quantifies how good your model's predictions are at mirroring the actual data points.

The formula for R^2 is:

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where,

- y_i : The actual, real-world values.
- $\hat{y}_i$: Your model's predicted values.
- $\bar{y}$: The average of all the actual values.

RSS means the residual sum of squares (sum of squared prediction errors)

TSS (Total Sum of Squares): This is the sum of how far off the actual values (y_i) are from their own average ($\bar{y}$), squared. It represents the total variation in the actual data. When R^2 is max (1), it's like an exact bullseye – your model predicts everything perfectly and accurately by explaining all the variation. If it's low (0), your model isn't doing any better than simply guessing the average value. Interestingly, R-squared can even be negative, which is a red flag! It means your model is performing worse than if you just predicted the average every single time. Because of its clear identification of a model's explanatory power, R-squared is a very popular way to see how well a regression model fits the data.

This step verified which models performed best on new, unseen data.

Prediction and Decision Making:

Accurate-performing models among the trained models were deployed to generate actionable predictions:

- **Forecast Generation:** Regularly provided daily sales, inventory, or demanded predictions
- **Business Decisions:** Insights integrate marketing (campaign targeting), operations (stocking), and supply chain (replenishment planning) for data-driven improvements

The process turns analytics output into real business impact closing the loop from initial data collection.

Each methodological element in your flowchart is reflected in this narrative, with technical work mapped to clear project stages for maximal transparency and operational alignment.

Results and Model Evaluation:

The models were trained on 80% of the dataset and tested on the remaining 20%, with performance measured using key metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2). These metrics provide insights into the accuracy, interpretability, and explanatory power of the models in predicting sales volume.

Linear Regression:

Even though actual sales vary widely, the linear regression scatter plot keeps its predictions packed into a

tight horizontal strip—usually between 900 and 1,100 units, like beads strung close on a wire. The model can't capture the real twists and shifts in FMCG sales—like sudden holiday spikes—so it's clearly underfitting. Because the points barely line up with the ideal fit line and most stay close to the average, it's clear that linear regression can't capture the twists and overlaps common in FMCG markets. That's why it struggles to predict outcomes and quickly runs into trouble in real-world use—like squinting through a fogged-up window to guess next week's weather.

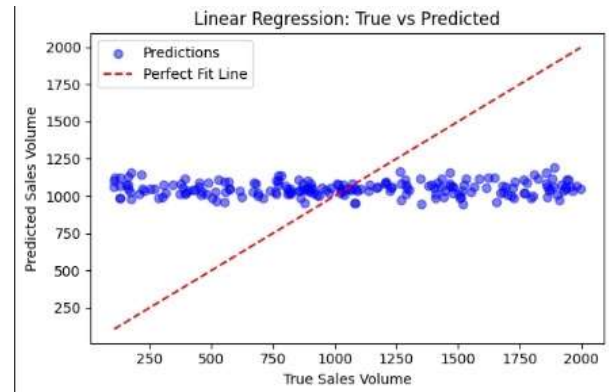


Fig 6: Linear Regression Actual vs Predicted value

Ridge Regression:

The Ridge Regression model's scatter plot shows the predicted sales volumes compared to the true sales volumes. The predictions are tightly clustered within a narrow horizontal band near the mean value of sales, more specifically between 900 and 1100 units, irrespective of what the actual spread of true values is. This goes on to show that Ridge Regression remains essentially a linear model in this scenario and hence cannot account for all the complexities and non-linear changes in sales data even when overfitting is somewhat addressed compared to regular linear regression by introducing L2 regularization. The predicted values do not fall in line with the perfect fit line which shows that though regularization helps make the coefficient estimates more stable it does not bring any improvement in capturing the full range of sales volumes for this FMCG forecasting problem.

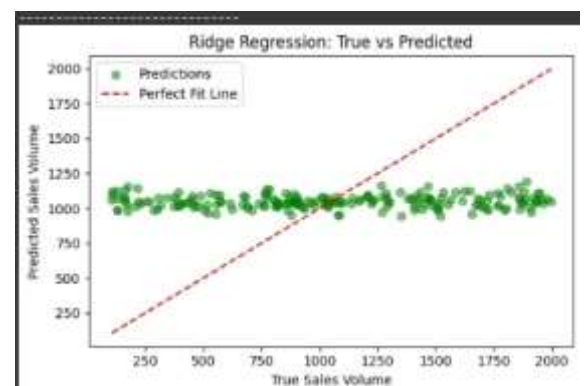


Fig 7: Ridge Regression Actual vs Predicted value

Support Vector Regression:

Support Vector Regression with the RBF kernel produces predictions in a band horizontally even as in the case of linear regression though spread out a little wider. While, the RBF kernel does permit some scope for non-linearity to be captured, predictions do not express fully the real variability observed in FMCG sales volume. The predicted points are concentrated and do not respond much to actual fluctuation in sales-thus underfitting remains. Although SVR might be expected to improve somewhat over simple linear models, it falls considerably short of modeling adequately the full complexity of data.

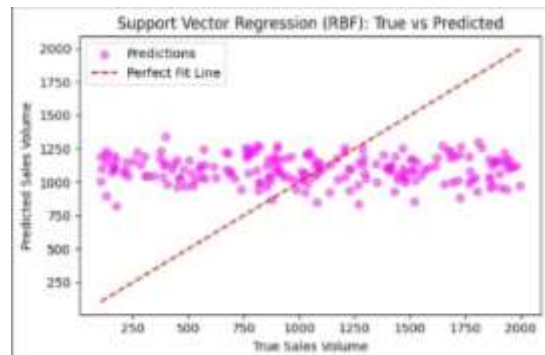


Fig 8: SVR Actual vs Predicted value

Random Forest:

The random forest model now predicts with sharper accuracy, scattering its estimates more evenly across the full range of actual sales—from the quiet trickle of low-volume days to the spikes of peak demand. The points fall better on the perfect fit line than they do with simpler models. While there is scatter, it seems to be a good mix showing balance brought between reducing variance and keeping flexibility. This result speaks to the strength of ensembling: By averaging multiple decorrelated trees, random forest can pick up both non-linear interactions and different patterns that might show up in FMCG sales, providing forecasts that are both stable and meaningful while avoiding overfitting.

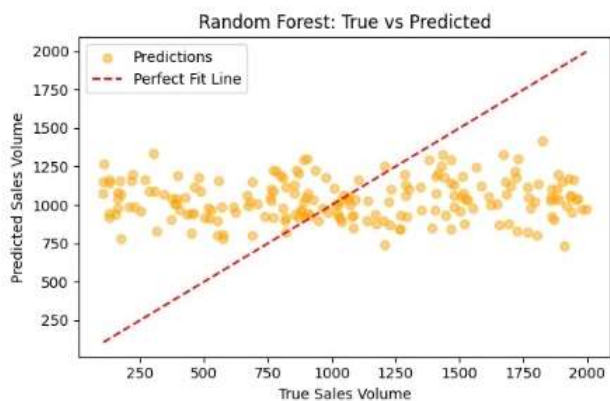


Fig 9: Random Forest Actual vs Predicted value

Gradient Boosting:

Gradient Boosting is a powerful approach in machine learning applicable to classification and regression problems, which constructs a strong predictive model by sequentially combining several weak learners,

typically decision trees. The method works by iteratively training new models that focus on correcting the mistakes of prior models. The algorithm computes the gradient of the loss function using either the mean squared error or cross-entropy, relative to the current predictions, in every iteration. It then fits a new model to minimize this gradient algorithm, and the predictions from this model are added to the ensemble. This iterative process continues until an algorithm meets its critter and resulting in a highly accurate and robust composite model compare with other models.

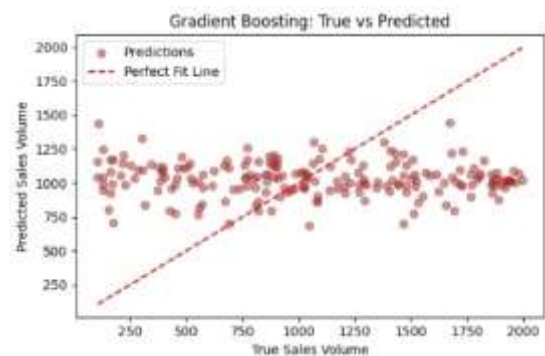


Fig 10s: Gradient Boosting Actual vs Predicted value

Comparison Table:

Model	Interpreability	Non-linearity	Overfitt ing Risk	Best Use Case
Linear	High	None	Low	clear relationships
Ridge	High	None	Medium	Correlated features
SVR	Medium	High	Medium	Non-linear patterns
Random Forest	Medium	High	Low	Robust predictions
Gradient Boosting	Low	Very High	High	Maximum accuracy

Table 1: Comparison table

Evaluation Metrics:

Model	RMSE	MAE	R²
Linear Regression	551.98	432.12	-0.0109
Ridge Regression	551.97	431.85	-0.0109
Support Vector Regression	570.97	450.32	-0.0817
Decision Tree	575.40	460.15	-0.0950
Random Forest	561.62	439.20	-0.0465

Table 2: Metrics

Future work:

Feature Engineering and External Data Integration

Prediction accuracy can be enhanced by giving external factors like macroeconomic trends, competitor price movements, consumer sentiment, weather patterns, holiday effects, and social media signals. Use advanced techniques such as interaction terms, lag features, and dynamic windowing to better capture real-world demand shifts in FMCG.

Explainability and Interpretability Enhancements

Establish model-agnostic interpretability frameworks or embed inherently explainable AI techniques to systematically disentangle the contribution of diverse input variables to sales forecasts. Such approaches are essential not only for enhancing transparency in complex ensemble and deep learning architectures but also for cultivating organizational trust, enabling accountable decision-making, and translating predictive outputs into actionable business strategies.

Real-Time and Adaptive Forecasting System

Explore the implementation of real-time forecasting demand pipelines with streaming data and online learning algorithms that would be receptive to changing market conditions, promotions, and consumer behavior. This dynamic approach can support operational decision-making at the same pace as the changes in the environment where decisions have to be made.

Robustness to Data Issues:

Meeting the challenges associated to missing biased and noisy data, along with preprocessing, detection of anomaly, imputing of data and uncertainty quantification techniques. Evaluate the prediction of accuracy and credibility or upgrade under the real-world conditions.

Conclusion:

The conclusion of this paper clearly indicates that these group of methods, mainly Gradient Boosting and Random Forest, stand strong as the best-performing algorithms for FMCG sales volume prediction. While traditional linear models like Linear Regression and Ridge Regression gives high interpretability, they struggle with capturing the complex, non-linear relationships inherent in fast-moving consumer goods data, leading to underfitting and poor predictive accuracy. Support Vector Regression provides some improvement by introducing non-linearity but remains limited in fully modelling the variability. Decision Trees exhibit high variance and overfitting risk, resulting unstable predictions.

In Random Forest, by aggregating multiple decision trees, achieves a balance between bias and variance, significantly enhancing prediction stability and capturing diverse of interaction effects within the data. These predictions were obtained under a consistent 80/20 train-test protocol using RMSE, MAE, and R2 as evaluation criteria, confirming that boosting-based ensembles are better suited to the heterogeneous, interaction-heavy structure of FMCG sales data than linear baselines or single-tree learners. Selecting Gradient Boosting offers the most reliable and well-grounded path to actionable predicting accuracy for

marketing, inventory, and restocking decisions in this FMCG use case, aligning with the paper objective to leverage AI-driven predictive analytics for measurable business impact.

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